

Lecture 11/Module 2

Transient Analysis of First and Second Order Circuit

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Today's Topics

- Series/Parallel connected Capacitors, Inductors
- Voltage, Current, Power across Capacitor, Inductor
- Initial Condition
- Order of a Circuit – First Order Circuit and Transient Analysis

Capacitors

A capacitor consists of two conducting plates separated by an insulator (or dielectric).

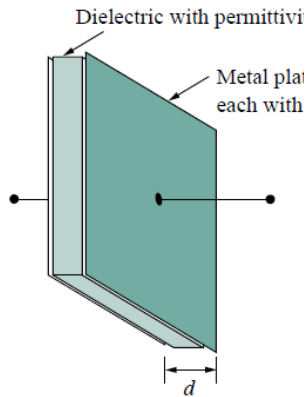


Figure 6.1 A typical capacitor.

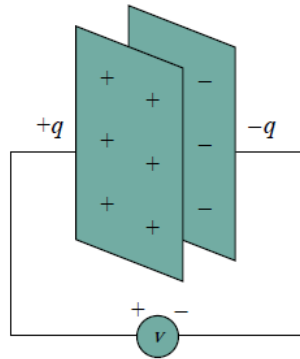


Figure 6.2 A capacitor with applied voltage v .

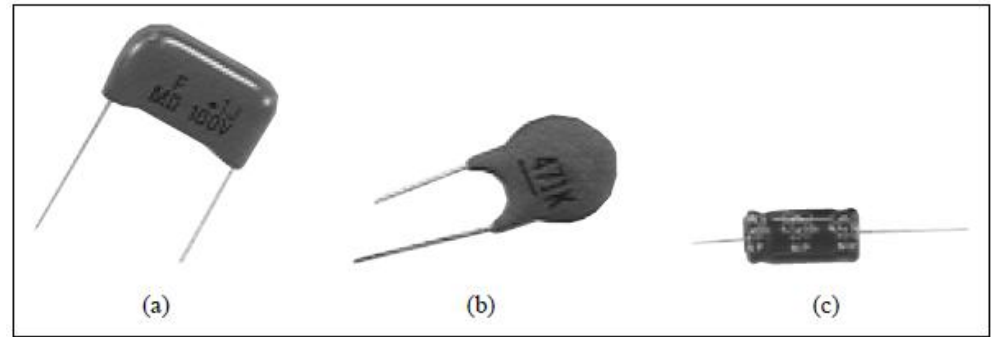


Figure 6.4 Fixed capacitors: (a) polyester capacitor, (b) ceramic capacitor, (c) electrolytic capacitor. (Courtesy of Tech America.)

Capacitor is an **energy storing** element

Capacitor opposes change in **voltage** & stores energy in **electric field**

Voltage, Current, Energy for Capacitors

The instantaneous power delivered to the capacitor is

$$p = vi = Cv \frac{dv}{dt} \quad (6.7)$$

The energy stored in the capacitor is therefore

$$w = \int_{-\infty}^t p \, dt = C \int_{-\infty}^t v \frac{dv}{dt} dt = C \int_{-\infty}^t v \, dv = \frac{1}{2} C v^2 \Big|_{t=-\infty}^t \quad (6.8)$$

We note that $v(-\infty) = 0$, because the capacitor was uncharged at $t = -\infty$. Thus,

$$w = \frac{1}{2} C v^2 \quad (6.9)$$

Using Eq. (6.1), we may rewrite Eq. (6.9) as

$$w = \frac{q^2}{2C} \quad (6.10)$$

$$v = \frac{1}{C} \int_{t_0}^t i \, dt + v(t_0)$$

Capacitance is the ratio of the charge on one plate of a capacitor to the voltage difference between the two plates, measured in farads (F).

Important Notes for Capacitors

A capacitor is an open circuit to dc.

The voltage on a capacitor cannot change abruptly.

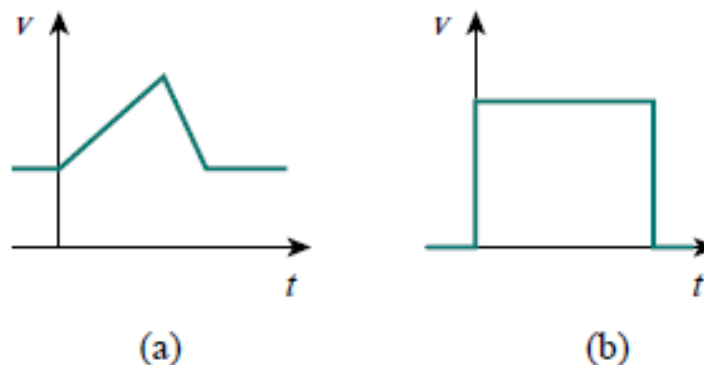
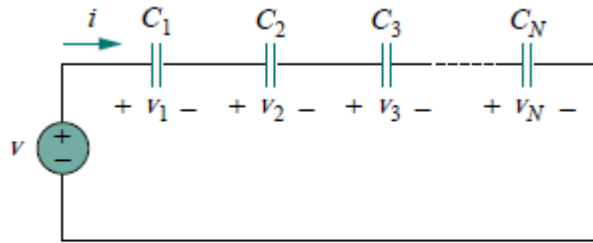
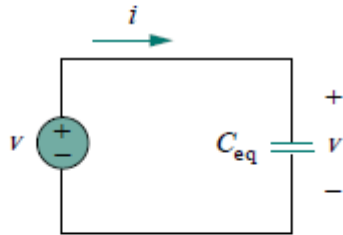


Figure 6.7 Voltage across a capacitor:
(a) allowed, (b) not allowable; an abrupt
change is not possible.

Series connected Capacitors



(a)

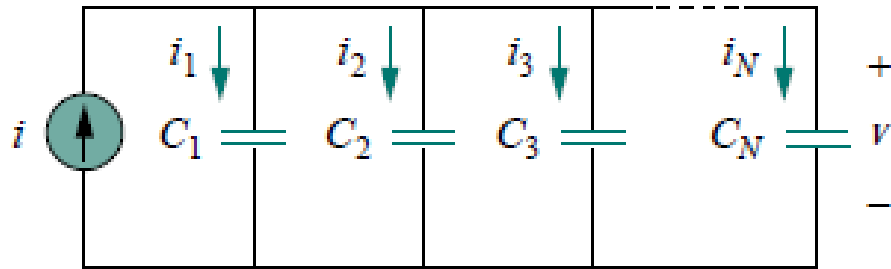


(b)

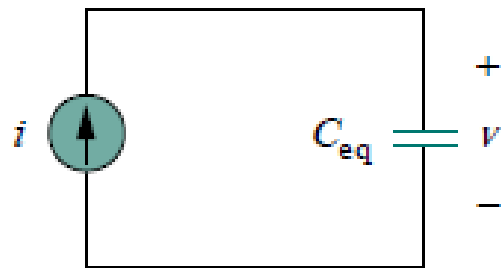
$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_N}$$

Figure 6.15 (a) Series-connected N capacitors, (b) equivalent circuit for the series capacitor.

Parallel connected Capacitors



(a)



(b)

$$C_{eq} = C_1 + C_2 + C_3 + \dots + C_N$$

Inductors

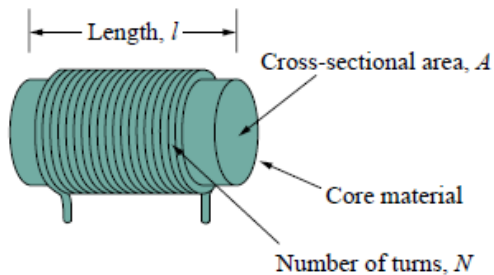
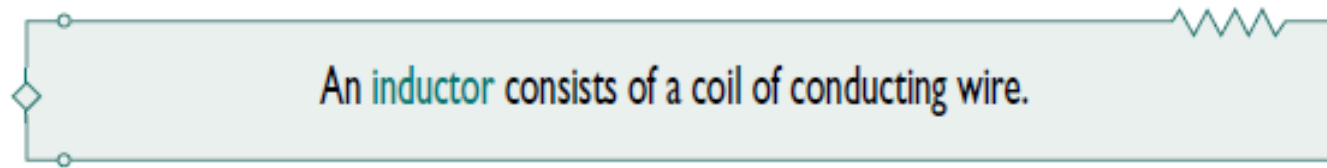


Figure 6.21 Typical form of an inductor.

Inductor is an **energy storing** element

Inductor opposes change in **current** & stores energy in **magnetic field**

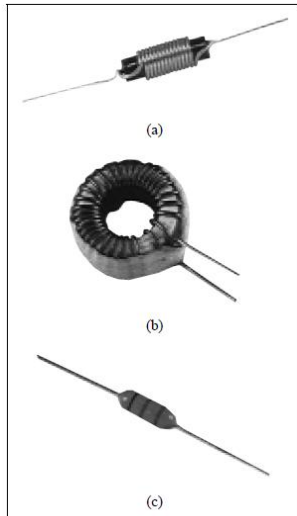


Figure 6.22 Various types of inductors:
(a) solenoidal wound inductor, (b) toroidal inductor, (c) chip inductor.
(Courtesy of Tech America.)

Voltage, Current, Energy for Inductors

Inductance is the property whereby an inductor exhibits opposition to the change of current flowing through it, measured in henrys (H).

$$v = L \frac{di}{dt}$$

$$w = \frac{1}{2} Li^2$$

$$w = \frac{1}{2} Li^2$$

$$i = \frac{1}{L} \int_{t_0}^t v(t) dt + i(t_0)$$

Important Notes for Inductors

An inductor acts like a short circuit to dc.

The current through an inductor cannot change instantaneously.

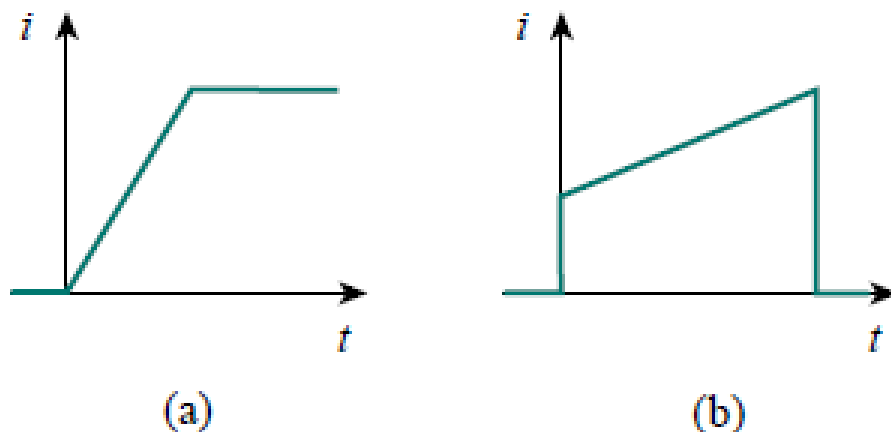
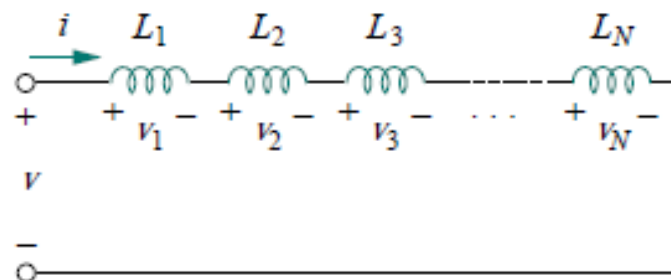


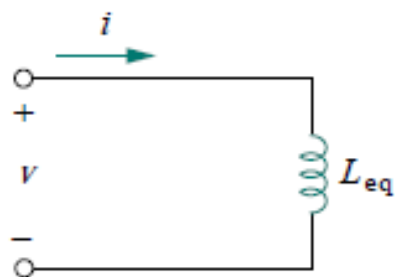
Figure 6.25 Current through an inductor: (a) allowed, (b) not allowable; an abrupt change is not possible.

Series connected Inductors



(a)

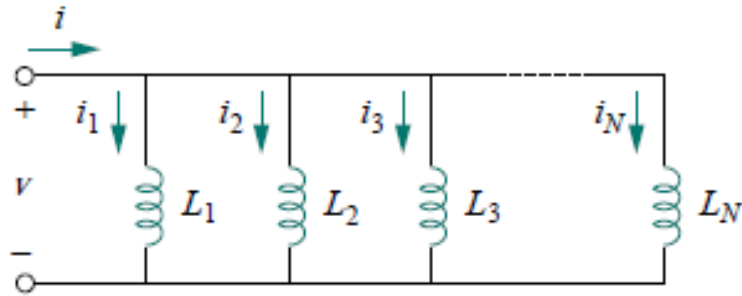
$$L_{\text{eq}} = L_1 + L_2 + L_3 + \dots + L_N$$



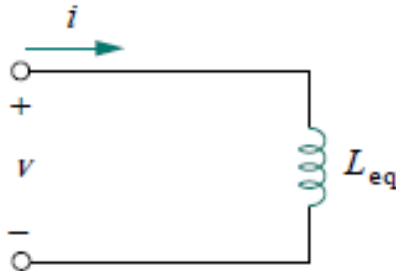
(b)

Figure 6.29 (a) A series connection of N inductors, (b) equivalent circuit for the series inductors.

Parallel connected Inductors



(a)



(b)

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N}$$

Figure 6.30 (a) A parallel connection of N inductors, (b) equivalent circuit for the parallel inductors.

Comparison between R, L and C

TABLE 6.1 Important characteristics of the basic elements.[†]

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v - i :	$v = iR$	$v = \frac{1}{C} \int_{t_0}^t i \, dt + v(t_0)$	$v = L \frac{di}{dt}$
i - v :	$i = v/R$	$i = C \frac{dv}{dt}$	$i = \frac{1}{L} \int_{t_0}^t v \, dt + i(t_0)$
p or w :	$p = i^2 R = \frac{v^2}{R}$	$w = \frac{1}{2} C v^2$	$w = \frac{1}{2} L i^2$
Series:	$R_{\text{eq}} = R_1 + R_2$	$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}$	$L_{\text{eq}} = L_1 + L_2$
Parallel:	$R_{\text{eq}} = \frac{R_1 R_2}{R_1 + R_2}$	$C_{\text{eq}} = C_1 + C_2$	$L_{\text{eq}} = \frac{L_1 L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

[†]Passive sign convention is assumed.

Order of a Circuit – First Order Circuit and Transient Analysis

Order of a circuit is determined by the highest order of Differential Equation describing the circuit

OR

Order of a circuit is determined by the number of independent storage element in a circuit

Examples

First-order circuit

An RC circuit has one independent capacitor:

$$RC \frac{dv_C}{dt} + v_C = v_s(t)$$

The highest derivative is first order → **first-order circuit**.

Similarly, a simple RL circuit is also first order.

Second-order circuit

An RLC circuit can have one independent capacitor and one independent inductor:

$$LC \frac{d^2v_C}{dt^2} + RC \frac{dv_C}{dt} + v_C = v_s(t)$$

The highest derivative is second order → **second-order circuit**.

Important Note

We carry out the analysis of RC and RL circuits by applying Kirchhoff's laws, as we did for resistive circuits. The only difference is that applying Kirchhoff's laws to purely resistive circuits results in algebraic equations, while applying the laws to RC and RL circuits produces differential equations, which are more difficult to solve than algebraic equations. The differential equations resulting from analyzing RC and RL circuits are of the first order. Hence, the circuits are collectively known as *first-order* circuits.

Transient analysis tells us what happens immediately after a change or switching event and before the circuit reaches its new steady state.

Transient analysis is the study of how a circuit behaves during the period when it is changing from one steady state to another.

Example

Suppose a capacitor is initially uncharged and we suddenly connect it to a DC voltage source through a resistor.

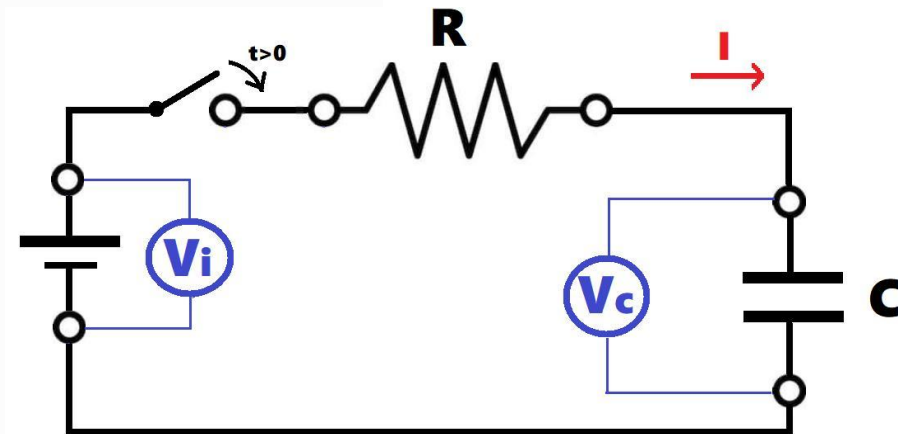
At $t = 0$, the switch is closed.

The capacitor voltage does **not** immediately jump to the source voltage. Instead,

$$v_C(t) = V \left(1 - e^{-t/RC} \right)$$

There are three stages:

1. **Before switching ($t < 0$)**
Capacitor is in its initial state.
2. **During the transition ($t > 0$)**
Capacitor voltage gradually changes.
This period is called the **transient period**.
3. **After a long time ($t \rightarrow \infty$)**
The capacitor reaches its final steady-state voltage.



Transient versus Steady State

Transient response	Steady-state response
Exists immediately after a change	Exists after transients die out
Changes with time	May become constant for DC circuits
Depends strongly on initial conditions	Depends on the final applied excitation
Usually contains exponential terms	Represents the final long-term behavior

For the RC example,

$$v_C(t) = \underbrace{V}_{\text{steady-state}} - \underbrace{Ve^{-t/RC}}_{\text{transient}}$$

So we can think of the total response as

Complete response = Transient response + Steady-state response
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Initial Condition

- For Transient Analysis it is important to know the initial condition of the circuit.
- We check the initial condition of a capacitive circuit by knowing the **initial voltage** across the **capacitor**
- We check the initial condition of an inductive circuit by knowing the **initial current** through the **inductor**

Thank You